

Nail Changes

Some cancer treatments can damage or change your fingernails and toenails. This might affect the appearance, texture, color, and health of your nails. Understanding these changes and how to manage them can help you keep your nails healthy during and after treatment.

- [Common nail changes during cancer treatment](#)
- [What cancer treatments cause nail changes?](#)
- [Nail changes by treatment type](#)
- [Other possible causes](#)
- [Treatments](#)
- [Tips for preventing and managing nail changes](#)
- [Learn more](#)

Common nail changes during cancer treatment

How your nails change during cancer treatment depends on what is causing the changes. Different treatments are more likely to cause certain types of changes. Depending on your treatment, you might notice changes in how your nails look or feel. You might also notice changes around or under your nails.

Common nail changes caused by cancer treatment include:

- **Weaker nails:** Nail thinning or reduced strength, leading to breakage
- **Painful nailbeds:** Delicate, sensitive nailbeds or cuticles
- **Dry or cracked skin:** Around your nails or cuticles
- **Color changes:** Yellowing, discoloration, or over-pigmentation on your nails or nailbeds
- **Irregular marks:** Ridges, pits, spots, or splitting of nails

- **Muehrcke lines:** Pairs of white lines across the nails that move as your nail grows out
- **Mees' lines:** Horizontal white line (usually only one) across the nails that doesn't move as your nail grows out
- **Beau's lines:** Horizontal ridges or grooves on nails
- **Nail plate issues:** Nail lifting or falling off (known as onycholysis)
- **Growth problems:** Slower growth than normal, or no growth at all
- **Infections:** Nail infections (known as paronychia), more common if nails are cracked or open

All of these changes can also be caused by non-cancer conditions, treatments, and medicines. It's important to tell your health care team if you have any nail symptoms, even if they aren't painful or bothering you.

What cancer treatments cause nail changes?

Some types of chemotherapy, immunotherapy, and targeted drug therapy can cause damage or changes to your nails. One, two, a few, or all of your nails could be affected.

Nail changes from chemo usually start sooner than changes caused by immunotherapy and targeted drug therapy. Most nail changes are temporary, but some can last long after treatment ends or can even be permanent.

Chemotherapy

Nail changes caused by chemo are usually short-term and return to normal within a few months of stopping treatment.

These types of chemo are more likely to cause nail changes:

- Vincristine (Oncovin, Vincasar PFS)
- Paclitaxel (Taxol)
- Fluorouracil (5FU, Carac)
- Etoposide (Vepesid, VP-16)
- Daunorubicin (Cerubidine)
- Bleomycin (Blenoxane)
- Cyclophosphamide (Cytoxan)
- Dacarbazine (DTIC-Dome)
- Methotrexate (Otrexup, Rasuvo, RediTrex, Trexall, Xatmep)

Immunotherapy

Most nail changes from [immunotherapy](#)¹ are caused by immune checkpoint inhibitors. Monoclonal antibodies are another type of immunotherapy that can cause nail changes, but this is less common.

Targeted drug therapy

Most nail changes from [targeted drug therapy](#)² are caused by EGFR inhibitors. VEGF (or angiogenesis inhibitors) are another type of targeted drug therapy that can also cause nail changes.

Nail changes by treatment type

Dark areas in cuticle

Cancer treatments that can cause dark areas in the cuticle include:

Targeted drug therapy antiangiogenic multikinase inhibitors

- sorafenib (Nexavar)
- sunitinib (Sutent)

Over-pigmented (hyperpigmented) nails

Cancer treatments that can cause hyperpigmented nails include:

Chemotherapy

- bleomycin (Blenoxane)
- capecitabine (Xeloda)
- cyclophosphamide (Cytoxan)
- dacarbazine (DTIC-Dome)
- daunorubicin (Cerubidine)
- doxorubicin (Adriamycin, Rubex)
- idarubicin (Idamycin)
- melphalan (Alkeran)

- methotrexate (Otrexup, Rasuvo, RediTrex, Trexall, Xatmep)

Targeted therapy, EGFR inhibitors

- erlotinib (Tarceva)
- gefitinib (Iressa)

Targeted therapy, monoclonal antibodies

- cetuximab (Erbix)
- panitumumab (Vectibix)

Fissures (deep grooves)

Cancer treatments that can cause fissures include:

Targeted drug therapy or immunotherapy monoclonal antibodies

- cetuximab (Erbix)
- panitumumab (Vectibix)

Hemorrhages (splinter hemorrhage)

Cancer treatments that can cause splinter hemorrhages include:

Chemotherapy

- doxorubicin (Adriamycin, Rubex)
- docetaxel (Taxotere)
- paclitaxel (Taxol)
- nab-paclitaxel (Abraxane)

Inflammation (paronychia)

Cancer treatments that can cause inflammation of the nail include:

Chemotherapy

- doxorubicin (Adriamycin, Rubex)
- docetaxel (Taxotere)
- paclitaxel (Taxol)
- nab-paclitaxel (Abraxane)

Targeted therapy, EGFR inhibitors

- erlotinib (Tarceva)
- gefitinib (Iressa)
- dacomitinib (Vizimipro)

Targeted therapy, monoclonal antibodies

- cetuximab (Erbix)
- necitumumab (Portrazza)
- panitumumab (Vectibix)

Targeted therapy, mTOR inhibitors

- everolimus (Afinitor, Votubia, Zortress)
- temsirolimus (Torisel)

Lifting of the nailbed (onycholysis)

Cancer treatments that can cause lifting of the nail bed include:

Chemotherapy

- dacarbazine (DTIC-Dome, DTIC)
- daunorubicin (Cerubidine, Vyxeos)
- mitoxantrone (Novantrone)

Targeted therapy, mTOR inhibitors

- everolimus (Afinitor, Votubia, Zortress)
- temsirolimus (Torisel)

Nail loss

Cancer treatments that can cause nail loss include:

Chemotherapy

- bleomycin (Blenoxane)
- 5-fluorouracil (5-FU, Adrucil)

Ridges, lines, creases, or other discoloration

Cancer treatments that can cause ridges, lines, creases, or other discoloration include:

Chemotherapy

- cyclophosphamide (Cytoxan, Neosar)
- doxorubicin (Adriamycin, Rubex)
- docetaxel (Taxotere)
- hydroxyurea (Hydrea)
- idarubicin (Idamycin)
- ifosfamide (Ifex)
- 5-fluorouracil (5-FU, Adrucil)

Other possible causes

There are a few other reasons your nails might change.

Certain cancers

Some cancers, especially skin cancers, can show up as nail changes. For example, some melanomas may appear as a thin, dark line or a bruise under the nail.

For pictures of melanomas under the nail, see [Skin Cancer Image Gallery](#)³.

Other medicines or treatments

Remember that while some cancers and cancer treatment can cause changes in your nails, non-cancer conditions and medications can also cause them. It's important to talk to your health care team about any medical problems you might have. Also make sure

they know all the medications, vitamins, minerals, and supplements you are taking.

With this information, you and your health care team can discuss your risk and what to expect.

Treatments

The treatment for nail changes depends on what signs or symptoms you're having. Not all nail changes need to be treated. Most changes are short-term and improve after cancer treatment ends.

Always talk to your doctor or cancer care team before starting or stopping any new medicine, supplement, or treatment.

For pain or swelling

While most nail changes aren't painful, nail inflammation or infection can cause pain or swelling. Your health care team might suggest daily vinegar nail soaks (equal parts white vinegar and water). Over-the-counter medicines such as non-steroidal anti-inflammatory drugs (NSAIDs) can also help relieve your symptoms.

For nail weakness, splitting, or ridges

For minor, non-painful nail changes like weakness, splitting, or ridges, you can use a water-soluble nail lacquer or a prescription nail polish to protect your nails from damage. Biotin (a dietary supplement) may also be prescribed to strengthen your nails.

For infected nails

If you have a nail infection, your health care team might prescribe antibiotics, antifungals, steroid ointment, NSAIDs, or other medicines.

For nailbed lifting (onycholysis)

If you have nailbed lifting with pain, you might need to have that nail removed surgically to relieve the pain.

Tips for preventing and managing nail changes

It may not be possible to prevent nail changes during cancer treatment, but there are

steps you can take to manage minor nail changes, and to avoid making them worse.

Protect your hands and feet:

- Check your hands and feet every day to look for changes in your nails.
- Avoid excessive or prolonged pressure, friction, or injury to your nails.
- Wear protective gloves or shoes during activities that could damage or injure your hands or feet.
- Wear socks and shoes that aren't too tight.

Use gentle soaps and skin-care products:

- Avoid skin-care products and lotions that have harsh chemicals, strong perfumes, or fragrances. These can be irritating.
- Wash your hands and nails with soap and water instead of hand sanitizer, which can sometimes irritate skin or nails.

Take extra care when trimming or polishing your nails:

- Keep your nails trimmed short, but don't cut or trim your cuticles (due to risk of infection).
- Don't get professional manicures or pedicures.
- Don't use nail polish removers or hardeners.
- Use a water-soluble nail lacquer or ask about a prescription nail polish if you have weak, brittle, or splitting nails.

Consider using cold packs if your chemo includes a taxane:

- If you are taking a taxane (especially docetaxel), consider applying cold packs to your nails for 15 minutes before the infusion, during the infusion, and for 15 minutes after the infusion. This may prevent nail lifting. Talk to your nurse or doctor first.

It's important to tell your health care team as soon as you notice any expected or unexpected changes to your nails. Preventing and avoiding injuries to your nails is an important part of keeping them as healthy as possible.

Learn more

- [Managing Cancer-related Side Effects](#)⁴
- [Hair, Skin, and Nail Changes](#)⁵

Hyperlinks

1. www.cancer.org/cancer/managing-cancer/treatment-types/immunotherapy.html
2. www.cancer.org/cancer/managing-cancer/treatment-types/targeted-therapy.html
3. www.cancer.org/cancer/types/skin-cancer/skin-cancer-image-gallery.html
4. www.cancer.org/cancer/managing-cancer/side-effects.html
5. www.cancer.org/cancer/managing-cancer/side-effects/hair-skin-nails.html

References

American Society of Clinical Oncology (ASCO). Skin conditions. Cancer.net. Content no longer available.

Emvalomati A, Ofliou V, Papageorgiou C, et al. Narrative Review of Drug-Associated Nail Toxicities in Oncologic Patients. *Dermatol Pract Concept*. 2023;13(1):e2023064. Accessed August 22, 2024. doi:10.5826/dpc.1301a64

Lacouture M, Sibaud V. Toxic side effects of targeted therapies and immunotherapies affecting the skin, oral mucosa, hair, and nails. *Am J Clin Dermatol*. 2018;19(1):31-39. Accessed August 22, 2024. doi:10.1007/s40257-018-0384-3

Rich P. Overview of nail disorders. *UpToDate*. UpToDate, Inc; 2024. Updated April 2024. Accessed August 22, 2024. <https://www.uptodate.com/contents/overview-of-nail-disorders>

Steele KT & Markova A. Cutaneous adverse events of molecularly targeted therapy and other biologic agents used for cancer therapy. *UpToDate*. UpToDate, Inc; 2024. Updated June 2024. Accessed August 22, 2024. <https://www.uptodate.com/contents/cutaneous-adverse-events-of-molecularly-targeted-therapy-and-other-biologic-agents-used-for-cancer-therapy>

Wetzel ML, Rubin AI, Hanania H, Patel AB. Treatment recommendations for nail unit toxicities secondary to targeted cancer therapy based on collective experience and

evidence-based literature review. *Am J Clin Dermatol*. 2021;87(1):180-183.
doi.org/10.1016/j.jaad.2021.07.022

Last Revised: September 12, 2024

Written by

American Cancer Society medical and editorial content team
(<https://www.cancer.org/cancer/acs-medical-content-and-news-staff.html>)

Developed by the with medical review and contribution by the American Society of Clinical Oncology (ASCO).

American Cancer Society medical information is copyrighted material. For reprint requests, please see our Content Usage Policy (www.cancer.org/about-us/policies/content-usage.html).

cancer.org | 1.800.227.2345